

Eliminating x gives $9y^2 + 9y - 18 = 0$, with solutions $y = -2$ and $y = 1$. The corresponding values of x are $x = 3$ and $x = 12$, so the critical points of f are $(3, -2)$ and $(12, 1)$. The discriminant is

$$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = (1)(18y + 18) - (-3)^2 = 18y + 9.$$

Since $D(3, -2) = -36 + 9 < 0$, we know that $(3, -2)$ is a saddle point of f . Since $D(12, 1) = 18 + 9 > 0$ and $f_{xx}(12, 1) = 1 > 0$, we know that $(12, 1)$ is a local minimum of f .

The second-derivative test does not give any information if $D = 0$. However, as the following example illustrates, we may still be able to classify the critical points.

Example 7 Classify the critical points of $f(x, y) = x^4 + y^4$, and $g(x, y) = -x^4 - y^4$, and $h(x, y) = x^4 - y^4$.

Solution Each of these functions has a critical point at $(0, 0)$. Since all the second partial derivatives are 0 there, each function has $D = 0$. Near the origin, the graphs of f , g and h look like the surfaces in Figures 15.13–15.15, respectively, so f has a local minimum at $(0, 0)$, and g has a local maximum at $(0, 0)$, and h is saddle-shaped at $(0, 0)$.

We can get the same results algebraically. Since $f(0, 0) = 0$ and $f(x, y) > 0$ elsewhere, f has a local minimum at the origin. Since $g(0, 0) = 0$ and $g(x, y) < 0$ elsewhere, g has a local maximum at the origin. Lastly, h is saddle-shaped at the origin since $h(0, 0) = 0$ and, away from the origin, $h(x, y) > 0$ on the x -axis and $h(x, y) < 0$ on the y -axis.

Exercises and Problems for Section 15.1

Exercises

- Figures (I)–(VI) show level curves of six functions around a critical point P . Does each function have a local maximum, a local minimum, or a saddle point at P ?
- Which of the points A, B, C in Figure 15.17 appear to be critical points? Classify those that are critical points.

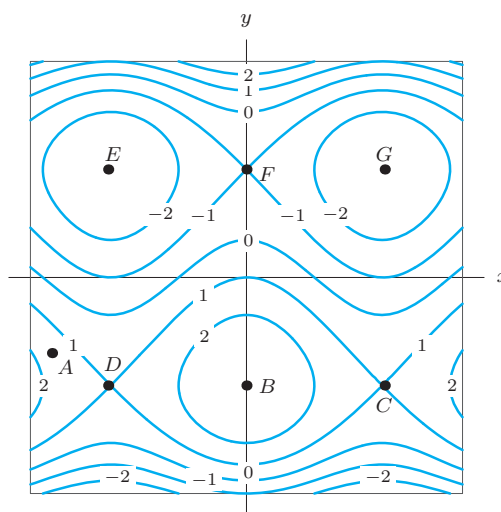
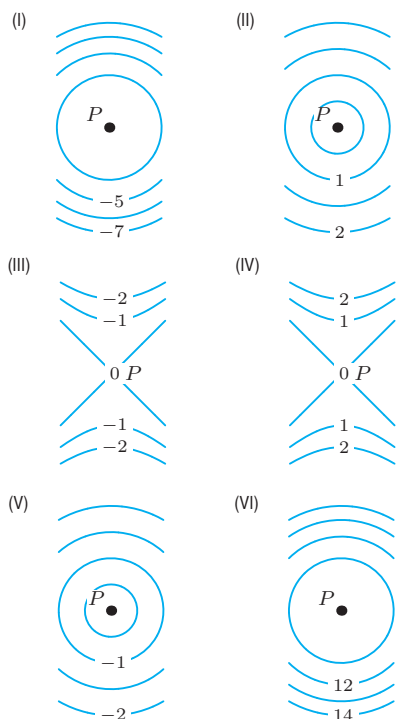


Figure 15.17

- Which of the points D – G in Figure 15.17 appear to be
 - Local maxima?
 - Local minima?
 - Saddle points?

Each function in Exercises 4–7 has a critical point at $(0, 0)$. What sort of critical point is it?

4. $g(x, y) = x^4 + y^3$ 5. $f(x, y) = x^6 + y^6$
 6. $k(x, y) = \sin x \sin y$ 7. $h(x, y) = \cos x \cos y$

For Exercises 8–20, find the critical points and classify them as local maxima, local minima, saddle points, or none of these.

8. $f(x, y) = x^2 - 2xy + 3y^2 - 8y$
 9. $f(x, y) = 5 + 6x - x^2 + xy - y^2$
 10. $f(x, y) = x^2 - y^2 + 4x + 2y$
 11. $f(x, y) = 400 - 3x^2 - 4x + 2xy - 5y^2 + 48y$
 12. $f(x, y) = 15 - x^2 + 2y^2 + 6x - 8y$
 13. $f(x, y) = x^2y + 2y^2 - 2xy + 6$
 14. $f(x, y) = 2x^3 - 3x^2y + 6x^2 - 6y^2$
 15. $f(x, y) = x^3 - 3x + y^3 - 3y$
 16. $f(x, y) = x^3 + y^3 - 3x^2 - 3y + 10$
 17. $f(x, y) = x^3 + y^3 - 6y^2 - 3x + 9$
 18. $f(x, y) = (x + y)(xy + 1)$
 19. $f(x, y) = 8xy - \frac{1}{4}(x + y)^4$
 20. $f(x, y) = e^{2x^2 + y^2}$

Problems

21. Find A and B so that $f(x, y) = x^2 + Ax + y^2 + B$ has a local minimum value of 20 at $(1, 0)$.
 22. For $f(x, y) = x^2 + xy + y^2 + ax + by + c$, find values of a , b , and c giving a local minimum at $(2, 5)$ and so that $f(2, 5) = 11$.
 23. (a) Find critical points for $f(x, y) = e^{-(x-a)^2 - (y-b)^2}$.
 (b) Find a and b such that the critical point is at $(-1, 5)$.
 (c) For the values of a and b in part (b), is $(-1, 5)$ a local maximum, local minimum, or a saddle point?
 24. Let $f(x, y) = kx^2 + y^2 - 4xy$. Determine the values of k (if any) for which the critical point at $(0, 0)$ is:
 (a) A saddle point
 (b) A local maximum
 (c) A local minimum
 25. Decide whether you think each point is a local maximum, local minimum, saddle point, or none of these.
 (a) P (b) Q (c) R (d) S
 26. Sketch the direction of ∇f at several points around each of P , Q , and R .
 27. At the points where $\|\nabla f\|$ is largest, put arrows showing the direction of ∇f .

For Problems 28–31, find critical points and classify them as local maxima, local minima, saddle points, or none of these.

28. $f(x, y) = x^3 + e^{-y^2}$
 29. $f(x, y) = \sin x \sin y$
 30. $f(x, y) = 1 - \cos x + y^2/2$
 31. $f(x, y) = e^x(1 - \cos y)$
 32. At the point $(1, 3)$, suppose that $f_x = f_y = 0$ and $f_{xx} > 0$, $f_{yy} > 0$, $f_{xy} = 0$.
 (a) What can you conclude about the behavior of the function near the point $(1, 3)$?
 (b) Sketch a possible contour diagram.
 33. At the point (a, b) , suppose that $f_x = f_y = 0$, $f_{xx} > 0$, $f_{yy} = 0$, $f_{xy} > 0$.
 (a) What can you conclude about the shape of the graph of f near the point (a, b) ?
 (b) Sketch a possible contour diagram.
 34. Draw a possible contour diagram of f such that $f_x(-1, 0) = 0$, $f_y(-1, 0) < 0$, $f_x(3, 3) > 0$, $f_y(3, 3) > 0$, and f has a local maximum at $(3, -3)$.
 35. Draw a possible contour diagram of a function with a saddle point at $(2, 1)$, a local minimum at $(2, 4)$, and no other critical points. Label the contours.

For Problems 25–27, use the contours of f in Figure 15.18.

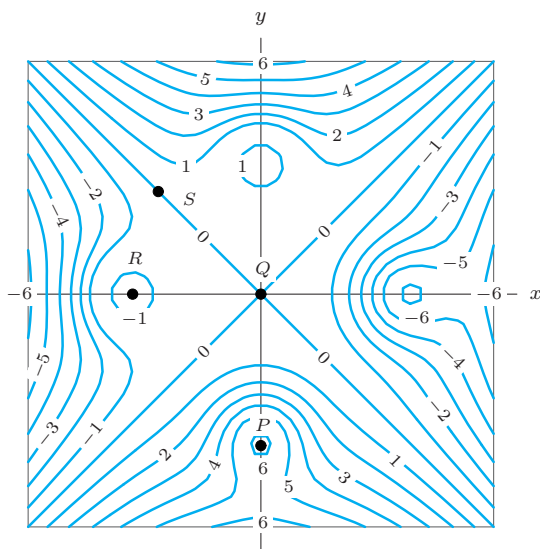


Figure 15.18

36. For constants a and b with $ab \neq 0$ and $ab \neq 1$, let

$$f(x, y) = ax^2 + by^2 - 2xy - 4x - 6y.$$

- (a) Find the x - and y -coordinates of the critical point. Your answer will be in terms of a and b .
 (b) If $a = b = 2$, is the critical point a local maximum, a local minimum, or neither? Give a reason for your answer.
 (c) Classify the critical point for all values of a and b with $ab \neq 0$ and $ab \neq 1$.
37. (a) Find the critical point of $f(x, y) = (x^2 - y)(x^2 + y)$.
 (b) Show that at the critical point, the discriminant $D = 0$, so the second-derivative test gives no information about the nature of the critical point.
 (c) Sketch contours near the critical point to determine whether it is a local maximum, a local minimum, a saddle point, or none of these.

38. On a computer, draw contour diagrams for functions

$$f(x, y) = k(x^2 + y^2) - 2xy$$

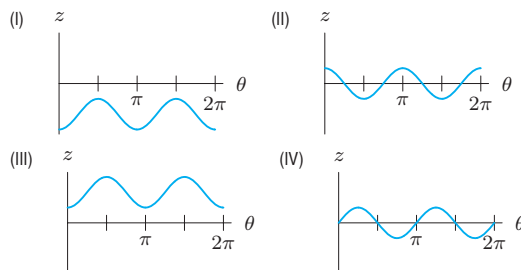
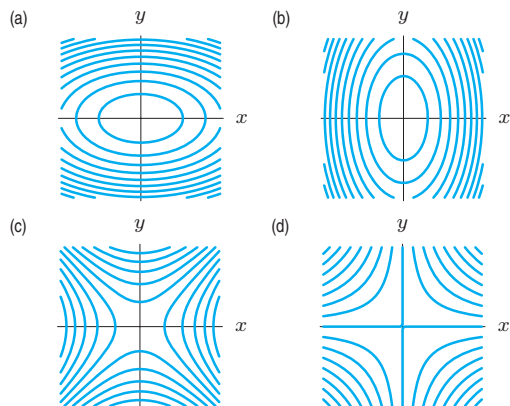
for $k = -2, -1, 0, 1, 2$. Use these figures to classify the critical point at $(0, 0)$ for each value of k . Explain your observations using the discriminant, D .

39. The behavior of a function can be complicated near a critical point where $D = 0$. Suppose that

$$f(x, y) = x^3 - 3xy^2.$$

Show that there is one critical point at $(0, 0)$ and that $D = 0$ there. Show that the contour for $f(x, y) = 0$ consists of three lines intersecting at the origin and that these lines divide the plane into six regions around the origin where f alternates from positive to negative. Sketch a contour diagram for f near $(0, 0)$. The graph of this function is called a *monkey saddle*.

40. The contour diagrams for four functions $z = f(x, y)$ are in (a)–(d). Each function has a critical point with $z = 0$ at the origin. Graphs (I)–(IV) show the value of z for these four functions on a small circle around the origin, expressed as function of θ , the angle between the positive x -axis and a line through the origin. Match the contour diagrams (a)–(d) with the graphs (I)–(IV). Classify the critical points as local maxima, local minima or saddle points.



Strengthen Your Understanding

In Problems 41–43, explain what is wrong with the statement.

41. If $f_x = f_y = 0$ at $(1, 3)$, then f has a local maximum or local minimum at $(1, 3)$.
 42. For $f(x, y)$, if $D = f_{xx}f_{yy} - (f_{xy})^2 = 0$ at (a, b) , then (a, b) is a saddle point.
 43. A critical point (a, b) for the function f must be a local minimum if both cross-sections for $x = a$ and $y = b$ are concave up.

In Problems 44–45, give an example of:

44. A nonlinear function having no critical points
 45. A function $f(x, y)$ with a local maximum at $(2, -3, 4)$.

Are the statements in Problems 46–56 true or false? Give reasons for your answer.

46. If $f_x(P_0) = f_y(P_0) = 0$, then P_0 is a critical point of f .
 47. If $f_x(P_0) = f_y(P_0) = 0$, then P_0 is a local maximum or local minimum of f .
 48. If P_0 is a critical point of f , then P_0 is either a local maximum or local minimum of f .
 49. If P_0 is a local maximum or local minimum of f , and not on the boundary of the domain of f , then P_0 is a critical point of f .
 50. The function $f(x, y) = \sqrt{x^2 + y^2}$ has a local minimum at the origin.
 51. The function $f(x, y) = x^2 - y^2$ has a local minimum at the origin.

52. If f has a local minimum at P_0 then so does the function $g(x, y) = f(x, y) + 5$.
53. If f has a local minimum at P_0 then the function $g(x, y) = -f(x, y)$ has a local maximum at P_0 .
54. Every function has at least one local maximum.
55. If P_0 is a local maximum of f , then $f(a, b) \leq f(P_0)$ for all points (a, b) in 2-space.
56. If P_0 is a local maximum of f , then P_0 is also a global maximum of f .

15.2 OPTIMIZATION

Suppose we want to find the highest and the lowest points in Colorado. A contour map is shown in Figure 15.19. The highest point is the top of a mountain peak (point A on the map, Mt. Elbert, 14,440 feet high). What about the lowest point? Colorado does not have large pits without drainage, like Death Valley in California. A drop of rain falling at any point in Colorado will eventually flow out of the state. If there is no local minimum inside the state, where is the lowest point? It must be on the state boundary at a point where a river is flowing out of the state (point B where the Arikaree River leaves the state, 3,315 feet high). The highest point in Colorado is a global maximum for the elevation function in Colorado and the lowest point is the global minimum.

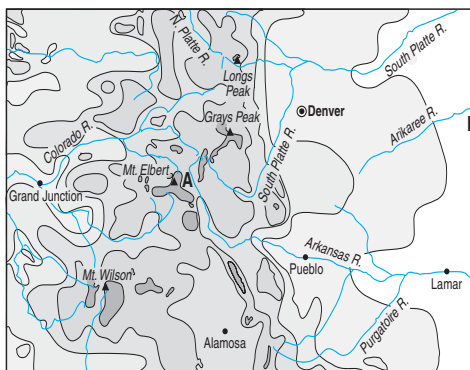


Figure 15.19: The highest and lowest points in the state of Colorado

In general, if we are given a function f defined on a region R , we say:

- f has a **global maximum on R** at the point P_0 if $f(P_0) \geq f(P)$ for all points P in R .
- f has a **global minimum on R** at the point P_0 if $f(P_0) \leq f(P)$ for all points P in R .

The process of finding a global maximum or minimum for a function f on a region R is called *optimization*. If the region R is not stated explicitly, we take it to be the whole xy -plane unless the context of the problem suggests otherwise.

How Do We Find Global Maxima and Minima?

As the Colorado example illustrates, a global extremum can occur either at a critical point inside the region or at a point on the boundary of the region. This is analogous to single-variable calculus, where a function achieves its global extrema on an interval either at a critical point inside the interval or at an endpoint of the interval.

To locate **global maxima and minima**:

- Find the critical points.
- Investigate whether the critical points give global maxima or minima.